JUAN MARCELO GUTIERREZ CARBALLO

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PROFESSIONAL SUMMARY

Ph.D. candidate specializing in cybersecurity with a 4.0 GPA, published research, and teaching experience across 10 graduate and undergraduate courses at Texas Tech, covering software security, communication networks, secure software design, and systems programming.

SKILLS

- ***Cybersecurity & Systems:*** *Computer Security, Network Security, Secure Software Design, Cryptography, Authentication, Access Control, Threat Modeling, Vulnerability Analysis.*
- ***Programming Languages:*** *C, C++, Python, Java, JavaScript, SQL, Assembly*
- ***Libraries & Frameworks:*** *Django, Node.js, jQuery*
- ***Database and Messaging:*** *MySQL, Microsoft SQL Server, Firebase, SQLite*
- ***Tools & Source Controls:*** *Git, Docker, Jira, SIEM tools, Intrusion Detection Systems (IDS).*

CERTIFICATIONS

Google Cybersecurity Professional Certificate 09/2025

- 9 hands-on, practice-based courses covering cybersecurity risk identification, threat analysis, vulnerability mitigation, and incident responses.

EXPERIENCE

Research Support Assistant, Beihang University Beijing, China, 2019 – 2021

- Explored GNU radio as a development toolkit for Software Defined Radio (SDR) ground stations, with attention to secure communication protocols and signal integrity.
- Supported research on the design of a micro-satellite ground station applying SDR as a concept to reduce hardware costs in the design.

Freelance Web Developer, Self-Employed 2018 – 2022

- Designed and developed web applications for small clients, applying security best practices including input validation, authentication flows, and protection against common vulnerabilities (XSS, CSRF, SQL injection).
- Built and maintained frontend interfaces using JavaScript and related frameworks; managed backend logic and database integration.
- Conducted code reviews and testing focused on identifying and resolving security vulnerabilities prior to deployment.

Math Lab Tutor, West Texas A&M University Canyon, Texas, 08/2016-05/2018

- Tutored undergraduate Computer Science students across the full range of Mathematics Department courses, adapting instruction to varying levels of difficulty and student background.

EDUCATION

Ph.D. in Computer Science 01/2023 - Present
Texas Tech University

- GPA: 4.0
- Research focus: Cybersecurity
- Recipient of the CS Ph.D. Fellowship since January 2023.

B.S. in Computer Science 08/2014 – 05/2018
West Texas A&M University

- Graduated with honors
- Dean's List 2014–2018
- Good Neighbor Scholarship recipient throughout undergraduate enrollment.

SELECTED PUBLICATIONS

Design of Secure Adaptable Connectors and State Machines for Software Architectures
Communications in Computer and Information Science (CCIS) — under review
Juan Marcelo Gutierrez Carballo, Michael Shin, Hassan Gomaa

A Software Product Line Approach for Design of Secure Software Architectural Patterns with Secure Connectors
Journal of Software: practice and experience — under review
Michael Shin, Hassan Gomaa, Taeghyun Kang, **Juan Marcelo Gutierrez Carballo**

Design of Adaptable and Secure Connectors for Software Architectures — 07/ 2024
Proceedings of the 19th International Conference on Software Technologies (ICSOFT 2024)
Juan Marcelo Gutierrez Carballo, Michael Shin, Hassan Gomaa

TEACHING ASSISTANTSHIP & LEADERSHIP EXPERIENCE

TTU CS5384 *Logic for Computer Scientists*	Fall 2024 - Present
TTU CS4331-02 *Programming Massively Parallel Processors*	Spring 2026
TTU CS5363 *Software Project Management*	Summer 2025
TTU CS5376 *Communication Networks*	Fall 2024
TTU CS5373 *Software Modeling and Architecture*	Summer 2024
TTU CS6378 *Software Security*	Spring 2024
TTU CS5332 *Software Engineering: Software Analytics*	Fall 2023
TTU CS5341 *Pattern Recognition*	Summer 2023
TTU CS1412 *Programming Principles*	Summer 2023
TTU CS4366 *Senior Capstone project*	Spring 2023

HONORS AND AWARDS

CS Ph.D. Fellowship, Texas Tech	Since 01/ 2023
MASTA Scholarship, Beihang University	09/2019-08/2020
Good Neighbor Scholarship, West Texas A&M University	08/2014-05/2018
Dean's List, West Texas A&M University	2014-2018